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Printed circuit board with electromagnetic bandgap structure for launcher in package devices

Abstract

A Launcher in Package (LiP) device for wireless communications, e.g., radar communications at 77 GHz, includes a printed circuit board (PCB) arranged between a radio frequency (RF) chip package and a radiating antenna structure. The PCB include an electromagnetic bandgap (EBG) structure including a plurality of EBG elements printed in a periodic pattern on the side of the PCB facing the radiating antenna structure. The plurality of EBG elements border at least one side of an RF channel through the PCB that couples an antenna element of the radiating antenna structure with a corresponding RF launcher in the RF chip package.

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Background/Summary

BACKGROUND

(1) Current trends in chips for wireless communication applications, including automotive and telecommunication technologies, are moving toward Launcher in Package (LiP) devices to improve millimeter wave (mmWave) performance and to reduce the overall printed circuit board (PCB) system cost. Generally, LiP devices are produced by stacking multiple components, such as an antenna structure and a PCB, on top of one another. However, the stacking of these components typically results in the formation of gaps, e.g., between the PCB and the antenna structure. For LiP devices that are configured to communicate via multiple radio frequency (RF) paths concurrently, such as in Multiple-Input Multiple-Output (MIMO) communications, these gaps lead to cross-

coupling between the internal RF channels of the LiP device which degrades the quality of the individual RF signals.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure may be better understood, and its numerous features and advantages made apparent to those skilled in the art by referencing the accompanying drawings. The use of the same reference symbols in different drawings indicates similar or identical items.
- (2) FIG. 1 shows a cross-sectional view of a Launcher in Package (LiP) device with an electromagnetic bandgap (EBG) structure on a printed circuit board (PCB) in accordance with some embodiments.
- (3) FIG. 2 shows a cross-sectional view of an EBG element of the plurality of EBG elements of an EBG structure in accordance with some embodiments.
- (4) FIG. 3 shows a Brillouin dispersion diagram showing a band gap for an EBG structure printed on a PCB with a gap of 100 micrometers (μm) to an antenna array structure in accordance with some embodiments.
- (5) FIGS. 4A-4B show additional Brillouin dispersion diagrams showing band gaps for an EBG structure printed on a PCB with varying gap heights between the PCB and the antenna array structure in accordance with some embodiments.
- (6) FIG. 5A shows an electric field illustration of a PCB in a LiP device without an EBG structure, and FIGS. 5B and 5C show electric field illustrations of a PCB in a LiP device with an EBG structure in accordance with some embodiments. The electric field in FIGS. 5A to 5C corresponds to the field for frequency signals at 77 gigahertz (GHz).
- (7) FIGS. 6A and 6B show diagrams of a symmetrical EBG configuration based on a second EBG structure. FIG. 6A shows a cross-sectional view of one of a plurality of second EBG elements, and FIG. 6B shows a cross-sectional view of a LiP device having a second EBG structure on the antenna structure facing the first EBG structure on the PCB in accordance with some embodiments.
- (8) FIG. 7 shows two Brillouin dispersion diagrams showing band gaps for the symmetrical EBG configuration shown in FIGS. 6A and 6B with varying gap heights between the PCB and the antenna structure in accordance with some embodiments.
- (9) FIG. 8A shows a layout diagram of an EBG structure printed on a PCB with respect to a radiating antenna element in accordance with some embodiments. FIGS. 8B and 8C show two electric field illustrations demonstrating the effect of the EBG structure shown in the layout diagram of FIG. 8A. FIG. 8B shows the propagation of the electric field without the EBG structure and FIG. 8C shows the isolation of the electric field with the EBG structure in accordance with some embodiments. The electric field in FIGS. 8B to 8C corresponds to an electric field for frequency signals at 77 GHz.
- (10) FIG. 9A shows a layout diagram of radiating antenna elements of an antenna structure without an EBG structure and FIG. 9B shows a corresponding graph illustrating the cross-coupling between RF channels associated with the radiating antenna elements of the antenna structure shown in FIG. 9A.
- (11) FIG. 10A shows a layout diagram of radiating antenna elements of an antenna structure with an EBG structure and FIG. 10B shows a corresponding graph illustrating the cross-coupling between RF channels associated with the radiating antenna elements of the antenna structure shown in FIG. 10A in accordance with some embodiments.
- (12) FIG. 11 shows a cross-sectional view of a LiP device with an EBG structure on a PCB facing a PCB fanout as the radiating antenna structure in accordance with some embodiments.
- (13) FIG. 12 shows a cross-sectional view of a LiP device with a symmetrical EBG configuration

having a second EBG structure, including a plurality of second EBG elements, on a PCB fanout as the radiating antenna structure facing the first EBG structure on the PCB in accordance with some embodiments.

DETAILED DESCRIPTION

(14) In a LiP device that communicates via multiple RF paths, the LiP device includes an antenna structure with a plurality of antenna elements (also referred to as radiating antenna elements or antenna array elements) attached to one side of a PCB and an RF integrated circuit (IC) chip package with a plurality of RF launchers (e.g., RF transmitters and RF receivers) attached to the other side of the PCB. Each of the RF launchers is communicatively coupled to a respective antenna element of the antenna structure via through-holes, metallic-filled vias, or other types of RF conductive strips in the PCB, thus forming an internal RF channel. During the process of manufacturing the LiP device, attaching the antenna structure to the PCB generally results in the formation of a gap between the PCB and the antenna structure. This gap reduces the isolation of the internal RF channels in the LiP device and results in increased cross-coupling between these internal RF channels. FIGS. 1-12 illustrate techniques and components for reducing the cross-coupling of internal RF channels in a LiP device by arranging an electromagnetic bandgap (EBG) structure on the PCB. The EBG structure provides a barrier that prevents or reduces energy from RF signals in one internal RF channel from leaking into adjacent internal RF channels, thereby increasing the transmission and reception signal quality of the LiP device.

(15) To illustrate by way of example, in one embodiment, the LiP device includes through-holes or metallic filled vias in the PCB that couple the RF launchers from the RF IC chip package with the corresponding transmission or reception antenna elements in the antenna structure. Each of these through-holes or metallic vias in the PCB corresponds to an internal RF channel. An EBG structure is printed on the surface of the PCB facing the antenna structure. For example, in one embodiment, the EBG structure includes a plurality of EBG elements arranged in a periodic pattern on the PCB and this plurality of EBG elements borders the through-holes or metallic vias in the PCB. The EBG structure reduces the RF energy leakage from one internal RF channel into adjacent internal RF channels, thereby decreasing the cross-coupling between the internal RF channels in the LiP device and increasing the transmission and reception signal quality. In addition, by printing the EBG structure directly on the PCB in the LiP device, the need to include an extra layer of milled EBG material is obviated.

(16) FIGS. 1-12 illustrate embodiments of an example of an EBG structure applied on a PCB in a LiP device to reduce the cross-coupling between internal RF channels in a LiP device. However, it will be appreciated that the apparatuses and techniques of the present disclosure are not limited to implementation in this particular LiP system, but instead may be implemented in any of a variety of LiP systems using the guidelines provided herein.

(17) FIG. 1 illustrates a cross-sectional view of a Launcher in Package (LiP) device **100** in accordance with some embodiments. In some embodiments, the LiP device **100** includes an electronic component assembly **120**, a PCB **102**, an antenna array structure **140**, an antenna protective enclosure **150**, and an EBG structure **108**, among other features. It is appreciated that in some embodiments, the LiP device **100** includes other components that may not be highlighted for purposes of conciseness and clarity of the ensuing explanation.

(18) The electronic component assembly **120** includes an RF IC chip package **122** including an RF IC chip **124** and RF launchers **126**, **128**. In some embodiments, the electronic component assembly **120** includes a heat sink **129** to transfer heat generated by the RF IC chip package **122** away from the RF IC chip package **122**. In some embodiments, interconnects **127** (only one instance labelled in the interest of clarity) adhere the RF IC chip package **122** to a first surface **104** of the PCB **102**. For example, interconnects **127** are solder balls of a ball grid array (BGA), pins of a pin grid array (PGA), interconnects of a land grid array (LGA), solder bumps, or other type of interconnect.

(19) The RF IC chip package **122** includes an RF IC chip **124** for conducting signal processing

operations in wireless communications. For example, in one embodiment, the RF IC chip is a radar chip for controlling radar operations in the 77 gigahertz (GHz) frequency band. Thus, the RF IC chip **124** can include low-power amplifiers, high-power amplifiers, modulators, demodulators, mixers, low-pass filters, analog-to-digital (ADC) converters, digital-to-analog (DAC) converters, as well as other components for processing received RF signals or for processing signals for RF transmission. In some embodiments, the RF IC chip **124** is coupled to one or more RF launchers such as RF launchers **126**, **128** through vias or interconnects in the RF IC chip package **122**. The RF launchers **126**, **128** are RF elements that transmit and/or receive electromagnetic signals. For example, the RF launchers **126**, **128** are radiative elements that are constructed from the metal layers of a chip interconnect stack or from redistribution layers of the chip. Although two RF launchers **126**, **128** are shown in FIG. 1, it is understood that this is for purposes of clarity and the number of RF launchers may be scalable to other quantities depending on the number of RF channels the LiP device **100** is designed to support. In this example, RF launcher **126** is a RF receiver to receive RF signals **126a** from other devices via a reception (Rx) antenna array **146** of the antenna structure **140** and RF launcher **128** is a RF transmitter to transmit RF signals **128a** from the LiP device **100** to other devices via a transmission (Tx) antenna array **148** of the antenna structure **140**. In some embodiments, the RF launchers **126**, **128** are arranged parallel to one another. In other embodiments, the RF launchers **126**, **128** are arranged orthogonal to one another.

(20) The PCB **102**, in some embodiments, is a flat laminate composite made from non-conducting substrate materials with layers of conductive materials, such as copper circuitry, located internally or in the external surfaces. The laminate layers, for example, include a flame retardant epoxy resin and glass fabric composite such as an FR4 material. The PCB **102** includes a first surface **104** facing the RF IC chip package **122**. For example, as shown in FIG. 1, the RF IC chip package **122** is attached to the first surface **104** of the PCB **102** with interconnects **127** made of any suitable solder material or alloy such as Tin, Silver, Copper, Aluminum, Nickel, Gallium, Germanium, Zinc, Silver, Gold, Bismuth, Indium, or any combination thereof. As shown, the attaching of the RF IC chip package **122** with the first surface **104** of the PCB **102** via the interconnects **127** results in a gap between the RF IC chip package **122** and the PCB **102**. This gap, for example, is in the range of 10-50 micrometers (μm).

(21) In some embodiments, the PCB **102** includes vias **112**. In some embodiments, the vias **112** are air-filled through-holes that are drilled or milled into the PCB **102** and serve as an RF channel to propagate RF signals between the RF launchers **126**, **128** and the corresponding antenna arrays in the antenna array structure **140**. For example, the left via **112** in the PCB **102** couples the RF launcher **128** serving as a RF transmitter with the Tx array **148** of the antenna structure **140**. In some embodiments, the vias **112** are metallic filled. For example, in the case of a substrate-integrated waveguide (SIW), the vias **112** are filled with a metallic material to form metallic vias.

(22) A second surface **106** of the PCB **102** faces the antenna structure **140**. The antenna structure **140**, in some embodiments, is a gap waveguide slot antenna with vias **144** in the antenna structure base layer **142** that are aligned with the vias **112** of the PCB **102**. To support multiple RF signals at the same time, e.g., to transmit and receive RF signals simultaneously, the antenna array structure **140**, in some embodiments, includes a plurality of antenna arrays **146**, **148**. In this example, for purposes of consistency with the RF launchers **126**, **128** in the RF IC chip package **122**, two antenna arrays are shown, but it is appreciated that the number of antenna arrays is scalable with the number of RF launchers in the RF IC chip package **122**. Furthermore, in the example shown in FIG. 1, antenna array **146** is a Rx antenna array to receive RF signals from other devices and forward them as RF signal **126a** to the RF launcher receiver **126**, and antenna array **148** is a Tx antenna array to receive RF signal **128a** from the RF launcher transmitter **128** and propagate this signal to other devices, i.e., external to LiP. In some embodiments, the antenna structure **140** is covered by an antenna protective enclosure **150**. For example, the antenna protective enclosure **150** is a weatherproof enclosure that protects the antenna structure **140** from external elements. For

example, the antenna protective enclosure **150** is a radome that is made of a thermoplastic material (e.g., a polyurethane- or polycarbonate-based material) that minimally attenuates the RF signal transmitted or received by the antenna structure **140** while protecting the antenna structure **140** from environmental factors such as humidity.

(23) As shown in FIG. 1, the LiP device **100** includes multiple internal RF channels **186**, **188**. A first internal RF channel **186** includes RF launcher **126** and antenna array **146** as well as the corresponding via **112** in the PCB **102** and via **144** in the antenna array structure **140** that couples them. A second internal RF channel **188** includes RF launcher **128** and antenna array **148** as well as the corresponding via **112** in the PCB **102** and via **144** in the antenna array structure **140** that couples them.

(24) In some embodiments, the process of stacking the PCB **102** and the antenna array structure **140** together (e.g., by screwing or soldering with soldering balls) results in the formation of a gap **107**. The gap **107**, for example, may be in the range of 100 μm to 300 μm and can lead to cross-coupling between the internal RF channels **186**, **188** of the LiP **100**. For example, the energy from RF signal **128a** in the internal RF channel **188** may leak into the adjacent internal RF channel **186** for RF signal **126a**. This cross-coupling of internal RF channels increases the amount of interference in the RF signal **126a**. In another example, if both RF launchers **126**, **128** are receivers, then a first RF signal received at a first antenna array and forwarded to a first RF receiver can leak into the RF channel corresponding to the reception of a second RF signal received at a second antenna array and forwarded to a second RF receiver. In order to reduce the cross-coupling of the internal RF channels, an EBG structure **108** is applied to the PCB in the gap **107** between the PCB **102** and the antenna structure **140**.

(25) In some embodiments, the EBG structure **108** is formed by printing a plurality of EBG elements in a periodic pattern on the PCB **102**. For example, in one embodiment, 3-D printing techniques are used to apply the EBG structure **108** including the plurality of EBG elements on the second surface **106** of the PCB **102**. As shown on the right-hand side of FIG. 1, in some embodiments, the EBG structure **108** is printed along the entire interface between the PCB **102** and the antenna array structure **140**. In other embodiments, as shown on the left-hand side of FIG. 1, the EBG structure **108** is printed along a border of the internal RF channel **188** created by the vias **112** and vias **144** of the PCB **102** and antenna array structure **140**, respectively. In either case, the EBG structure **108** lowers the cross-coupling between the internal RF channels by acting as a barrier for energy leakage attributed to RF signals at specific frequencies, e.g., at 77 GHz.

(26) In some embodiments, the geometry of the EBG elements and the periodic pattern for the EBG structure is developed utilizing a model analysis based on Brillouin dispersion diagrams. Example geometric configurations for periodic EBG structures printed on a PCB are described in the following figures.

(27) FIG. 2 illustrates an example of a cross-sectional view **200** taken from FIG. 1 focused on an individual EBG element **202** of the plurality of EBG elements in the EBG structure **108**. A section of the antenna array structure **140** and a gap **207** are also shown. In some embodiments, gap **207** corresponds, at least in part, to gap **107** in FIG. 1. In the cross-sectional view **200**, the antenna array structure **140** is shown above the EBG element **202**; that is, the cross-sectional view **200** is rotated 180 degrees relative to the cross-sectional view shown in FIG. 1.

(28) In some embodiments, the EBG element **202** includes a supporting column **206** and a top portion **204** centered on top of the supporting column **202**. In some embodiments, the supporting column **206** and the top portion **204** are made of the same EBG material. While the explanation herein describes the top portion **204** of the EBG element **202** as a disc for purposes of conciseness and clarity, it is appreciated that other shapes (e.g., quadrilaterals, octagons, dodecagons, or other polygons with a higher number of sides) can also be used. For example, instead of being in disc form as described herein, top portion **204** can be in the shape of an octagon. In some embodiments, the height **210** of the supporting column **206** is between 150 μm and 350 μm . In some

embodiments, the diameter **206d** of the supporting column **206** is between 150 μm and 500 μm . In some embodiments, the thickness **212** of the top portion **204** is between 15 μm and 75 μm . In some embodiments, the diameter **204d** (or width across the widest part of the polygon in the case of another polygon shape) of the top portion **204** is between 500 μm and 900 μm . In some embodiments, a ratio of the diameter **204d** of the top portion **204** to the diameter **206d** of the supporting column **206** is between 1:1 and 5:1. For example, the diameter **204d** of the top portion **204** is 700 μm and the diameter **206d** of the supporting column **206** is 250 μm . In some embodiments, the supporting column **206** is printed on top of a metallization layer **208**. For example, the metallization layer has a thickness **214** between 15 μm and 100 μm . In some embodiments, referring to the LiP device **100** in FIG. 1, the metallization layer **208** is printed on top of the second surface **106** of the PCB **102**. In some embodiments, the height of the gap **207** is between 0 μm and 500 μm , e.g., between 50 μm and 250 μm .

(29) In some embodiments, the EBG structure **108**, including the metallization layer **208** and the plurality EBG elements **202**, is designed utilizing a model analysis based on Brillouin dispersion diagrams. This Brillouin dispersion diagram modeling analysis is performed for different geometries for the plurality of EBG elements and different periodic patterns of the plurality of EBG elements to determine the EBG structure configuration on the PCB that provides a band gap at the desired frequency, e.g., at 77 GHz. For example, the EBG structure **108** is designed so the plurality of EBG elements (e.g., corresponding to multiple instances of EBG element **202**) form a periodic pattern with regular interval spacings between them. In some embodiments, the periodic pattern is achieved by repeating the structure side by side in the two dimensions such as that shown in FIG. 8A, for example. In some embodiments, the distance between adjacent EBG elements **202** (as measured from the outer edges of respective top portions **204**) ranges from 50 nm to 250 nm. In some embodiments, referring to FIG. 2, distance **218** is between 25 nm and 125 nm. For example, the distance between adjacent EBG elements **202** is 100 nm, i.e., distance **218** is 50 nm. Based on this design, the EBG structure is applied to the PCB utilizing 3-D printing methods, microlithography methods, or other similar printing methods. Referring to FIG. 2, in some embodiments, the antenna structure **140** is a metal, metallized plastic, or another PCB with a copper layer. In some embodiments, the gap **207** is filled with air or a dielectric material. In other embodiments, the gap **207** is a vacuum. In some embodiments, the material of the EBG element **202** and/or the material of the base layer **208** includes a metal such as copper. The copper, in some embodiments, can be plated such as with an ENIG (Electroless nickel immersion gold) or an ENEPIG (Electroless nickel electroless palladium immersion gold) plating. In some embodiments, the filling **220** adjacent to and in between the plurality of EBG elements **202** is a dielectric or PCB material.

(30) FIG. 3 illustrates an example of a Brillouin dispersion diagram **310** corresponding to an EBG structure **108** with a plurality of EBG elements **202** printed on a PCB. The x-axis of the Brillouin dispersion diagram **310** is the wave vector of the Brillouin zone of the EBG elements in the EBG structure **108** and the y-axis represents the frequency (in GHz). Lines **312** and **314** correspond to a first mode (**1** mode) and a second mode (**2** mode) of a parameter sweep across the Brillouin Zone using periodic boundary conditions. The area marked by **316** represents the band gap and falls between 60 GHz and 100 GHz. As used herein, the term “band gap” indicates a range at which frequencies do not propagate through the EBG structure. In other words, the EBG structure operates as a barrier to frequencies falling in the band gap. The Brillouin dispersion diagram **310** demonstrates results corresponding to an EBG structure **108** printed on a PCB with each of the plurality of EBG elements **202** designed according to the following dimensions (with reference made to the dimensions discussed above in FIG. 2).

(31) The gap height **207** between the EBG structure **108** and the antenna structure **140** is 100 μm . The EBG element **202** includes a supporting column **206** with a height **210** of 254 μm and a diameter **206d** of 250 μm . The EBG element **202** also includes a top part **204** in the form of a disc

with a diameter **204d** of 700 μm and a thickness **212** of 35 μm . The thickness **214** of metallization **208** is 35 μm . As shown in Brillouin dispersion diagram **310**, this EBG element design results in a band gap from 60 GHz to 100 GHz. Accordingly, for a LiP device conducting radar communications at 77 GHz, the EBG structure provides a barrier so that energy associated with signals at these frequencies do not leak into other internal RF channels of the LiP device. This in turn reduces or eliminates the cross-coupling of an internal RF channel transmitting and/or receiving signals at 77 GHz with other internal RF channels of the LiP device.

(32) FIGS. **4A-4B** illustrates a series of additional Brillouin dispersion diagrams **410**, **420**, **430**, **440** for a EBG structure design similar to one described above in FIG. **3** with varying gap heights, i.e., corresponding to gap height **207**. In Brillouin dispersion diagram **410**, the gap height **207** is zero. This results in a band gap **416** between 10 GHz (the starting point of the analysis) to 100 GHz. In Brillouin dispersion diagram **420**, the gap height **207** is 1 μm . This results in a band gap **426** between 10 GHz and 100 GHz. In Brillouin dispersion diagram **430**, the gap height **207** is 200 μm , resulting in a band gap **436** between 65 GHz and 106 GHz. In Brillouin dispersion diagram **440**, the gap height **207** is increased to 300 μm . This results in a band gap **446** between 68 GHz and 104 GHz. Accordingly, for radar communications at 77 GHz, the EBG structure with a plurality of EBG elements with the dimensions described above provides a barrier to energies at 77 GHz and serves to isolate an internal RF channel carrying signals at 77 GHz for a variety of different gap heights in a LiP device. Accordingly, cross-coupling with other internal RF channels is greatly reduced or eliminated altogether.

(33) FIGS. **5A-5C** shows a series of illustrations demonstrating the electric field in a PCB generated by an RF signal at 77 GHz being propagated through an internal RF channel. FIG. **5A** shows the corresponding electric field in a PCB without an EBG structure, and FIGS. **5B-5C** show the corresponding electric field in a PCB with an EBG structure.

(34) In illustration **510** of FIG. **5A**, the electric field in the PCB is shown for an RF signal with a frequency of 77 GHz being transmitted via an internal RF channel (e.g., via a through-hole or metallic via) in the PCB without the presence of an EBG structure. As shown, the energy from the RF signal propagates in all directions **516** away from the internal RF channel **512** (i.e., the energy leaks out over the area covered by the PCB). One example of an area with a high amount of energy leakage is shown at area **514**. This energy leakage to other areas of the PCB causes cross-coupling with adjacent internal RF channels (not shown) in the PCB.

(35) In illustration **520** of FIG. **5B**, an EBG structure with a plurality of EBG elements (two instances indicated by **524**) is arranged on the PCB and borders the internal RF channel **522** transmitting the RF signal at 77 GHz. As shown, the energy generated by the RF signal at 77 GHz is contained by the EBG structure **524**. The barrier created by the EBG structure is shown by box **526**. In this manner, the EBG structure **524** prevents leakage of the energy of the RF signal at 77 GHz from internal RF channel **522** into other internal RF channels (not shown).

(36) In illustration **530** of FIG. **5C**, the EBG structure including a plurality of EBG elements **540** (only one labelled in the interest of clarity) is depicted to illustrate the containment of the energy from the RF signal at 77 GHz to the internal RF channel **532**. As shown, the two adjacent internal RF channels **534**, **536** are mostly free from energy leaking into them from internal RF channel **532**. In this manner, the EBG structure including the plurality of EBG elements **540** acts as a barrier to contain the energy from the RF signal at 77 GHz to the internal RF channel **532**.

(37) FIG. **6A** illustrates a perspective view **600** and cross-sectional view **650** of a symmetrical EBG configuration according to some embodiments. In this configuration, a second EBG structure with a plurality of EBG elements **640** (one shown) is applied to a side of the antenna structure facing the PCB. For example, referring to FIG. **1**, the second EBG structure is applied to a side of antenna structure **140** in the gap **107** and facing EBG **108**. In some embodiments, each of the plurality of the EBG elements **640** of the second EBG structure is aligned with an EBG element **202** of the EBG structure **108** on the PCB. In some embodiments, each of plurality of EBG elements **640** in

the second EBG structure has a base portion **602** and a protrusion portion **604** extending from the base portion **702** in the direction of each of the EBG elements **202** of the EBG structure **108**.

(38) For example, each EBG element **640** in the second EBG structure includes a base portion **602** and a protrusion portion **604**. The base portion **602** has a diameter **602d** in the range of 500 μm to 1000 μm , and the protrusion portion **604** has a diameter **604d** in the range of 250 μm to 750 μm and a height **604h** in the range of 500 μm to 1000 μm . In some embodiments, the dimensions of the EBG elements **640** in the second EBG structure on the antenna side **140** are designed based on the dimensions of the EBG elements **602** in EBG structure **108** on the PCB **102**. To illustrate by way of example, if each EBG element **202** in the EBG structure **108** on the PCB **102** side has a supporting column width of 400 μm , a supporting column height of 250 μm , a top portion in the shape of a disc with a diameter of 700 μm , and a disc thickness of 35 μm , each EBG element **640** in the second EBG structure on the antenna side **140** has a protrusion portion diameter **604d** of 500 μm , a protrusion portion height **604h** of 800 μm , and a base portion diameter **602d** of 800 μm . The gap **607**, for example, is 100 μm . In some embodiments, gap **607** corresponds, at least in part, to gap **107** in FIG. 1.

(39) FIG. 6B illustrates a cross-sectional view of a LiP device **660** in accordance with various embodiments. In some embodiments, the LiP device **660** corresponds substantially to the LiP device **100** illustrated in FIG. 1 with a difference that a second EBG structure **608** is included on the antenna structure **140**. The second EBG structure **608** includes a plurality of second EBG elements such as EBG elements **640** shown in FIG. 6A. The dashed line box **650** corresponds to diagram **650** in FIG. 6A. In some embodiments, the second EBG structure **608** is applied to the antenna structure **140** according to one of various techniques. If the antenna structure **140** is a metal, then it is subject to a milling process to produce the second EBG structure **608**. If the antenna structure **140** is a metallized plastic, then the second EBG structure **608** formed in the injection mold when the antenna structure **140** is formed. In another embodiment, if the antenna structure is a PCB fanout (such as that shown in FIGS. 11 and 12), then the second EBG structure **608** is formed during the PCB manufacturing of the PCB fanout.

(40) By implementing a symmetrical EBG configuration on both sides of the gap **607** as shown in FIG. 6A-6B, the band gap of the associated symmetrical EBG configuration can be more focused on a particular frequency. For example, FIG. 7 shows two Brillouin dispersion diagrams **710** and **720**. Brillouin dispersion diagram **710** corresponds to a symmetrical EBG configuration as shown in FIG. 6A with a gap height **607** of 100 μm . Brillouin dispersion diagram **720** corresponds to a symmetrical EBG configuration as shown in FIG. 6A with a gap height **607** of 200 μm . The band gap **716** in Brillouin dispersion diagram **710** is at 65 GHz to 90 GHz. The band gap **726** in Brillouin dispersion diagram **720** is at 72 GHz to 86 GHz. Accordingly, the symmetrical EBG configuration is more frequency-selective since its corresponding band gaps cover a narrower frequency range compared to the band gaps attributed to the single EBG structure on only the PCB side (e.g., as shown in Brillouin dispersion diagram **410** for a gap height of 100 μm or by Brillouin dispersion diagram **530** for a gap height of 200 μm).

(41) FIG. 8A shows a schematic diagram **810** of an antenna structure side perspective view of a radiating antenna element **812** of an antenna array structure **814** arranged over a PCB **816** with an EBG structure including a plurality of EBG elements **818** (only one labelled in the interest of clarity) arranged thereon. In this example, the EBG structure including the plurality of EBG elements **818** on the PCB **816** is arranged to surround the radiating antenna element **812** of an antenna array structure **814**. FIG. 8B and FIG. 8C depict the electric field in the gap between the PCB **816** and the antenna structure **814** without the EBG structure (illustration **820** in FIG. 8B) and with the EBG structure (illustration **830** in FIG. 8C). The electric field corresponds to a frequency signal at 77 GHz.

(42) In some embodiments, for schematic diagram shown in FIG. 8A, the PCB **816** corresponds to PCB **102** in FIG. 1, the antenna array structure **814** corresponds to antenna array structure **140** in

FIG. 1 or 6 (or PCB fanout **1140** in FIG. 11 and PCB fanout **1240** in FIG. 12), radiating antenna element **812** corresponds to Tx antenna array **148** in FIG. 1 or 6 (or portions of PCB fanout **1140** in FIG. 11 and PCB fanout **1240** in FIG. 12), and the EBG structure with the plurality of EBG elements **818** corresponds to EBG structure **108** in FIG. 1, 6, 11, or 12, where each of the EBG elements **818** corresponds to an EBG element **202** such as the one shown in FIG. 2. From the antenna side perspective view of schematic diagram **810**, the RF launcher in the RF IC package corresponding to the radiating antenna element **812** is not shown but is coupled with the radiating antenna element **812** via a corresponding internal RF channel through PCB **816** as described herein.

(43) FIG. 8B shows an illustration of the electric field at 77 GHz in the gap between the PCB **816** and the antenna structure **814** without an EBG structure. As shown, the electric field for the radiating antenna element **822** operating at 77 GHz propagates (shown by arrows **826**) throughout the entire area of the gap. This can lead to cross-coupling with other internal RF channels (not shown).

(44) FIG. 8C shows an illustration of the electric field at 77 GHz in the gap between the PCB **816** and the antenna structure **814** with an EBG structure including a plurality of EBG elements **818** (only one labelled in the interest of clarity). As seen in illustration **830**, the electric field is contained to the area **836** corresponding to the footprint of the radiating antenna element **812**, i.e., the EBG structure with the plurality of EBG elements **818** creates a barrier so that the electric field of the RF signal at 77 GHz does not propagate over the entire area of the gap as shown in FIG. 8B.

(45) FIGS. 9A-9B and 10A-10B illustrate a comparison of the cross-coupling in internal RF channels of a LiP device without an EBG structure (FIG. 9A-9B) and a LiP device with an EBG structure (FIG. 10A-10B) according to some embodiments. FIG. 9A illustrates a layout diagram without an EBG structure **910**, and FIG. 10A illustrates a layout diagram with an EBG structure **1010**. Each of layout diagrams **910** and **1010** include four radiating antenna elements (**912a-912d** in layout diagram **910** and **1012a-1012d** in layout diagram **1010**), with each radiating antenna element corresponding to an internal RF channel in the LiP device. FIG. 9B shows a graph **920** corresponding to FIG. 9A, and FIG. 10B shows a graph **1020** corresponding to FIG. 10A. Both of graphs **920**, **1020** illustrate the cross-coupling between the different internal RF channels corresponding to the radiating antenna elements. It is appreciated that the specific configuration of radiating antenna elements shown in layout diagrams **910**, **1010** are illustrative in nature and other configurations are similarly considered. For example, while four radiating antenna elements are shown, it is appreciated that this number is scalable to other quantities depending on the number of RF channels the LiP device is designed to support.

(46) In FIG. 9A, the layout diagram **910** shows a view of the antenna structure **912** with four radiating antenna elements **912a-912d**. Each of the radiating antenna elements are coupled to an RF launcher in the RF IC chip package via a corresponding internal RF channel of the LiP device as described herein. In this example, the PCB and RF launchers on the RF IC chip package, as well as the corresponding internal RF channel linking the radiating antenna elements to the RF launchers, are not shown for purposes of illustrating the configuration of the radiating antenna elements **912a-912d**.

(47) As shown, the first radiating antenna element **912a** and the third radiating antenna element **912c** are parallel to one another and orthogonal to both the second radiating antenna element **912b** and the fourth radiating antenna element **912d**. Graph **920** in FIG. 9B shows the coupling between the internal RF channels associated with the radiating antenna elements **912a-912d**. The x-axis of graph **920** is the frequency range (in GHz) and the y-axis shows the cross-coupling between the internal RF channels associated with the radiating antenna elements **912a-912d** of the layout diagram **910** in dB. The lower the dB (i.e., the more negative), the lower the coupling between the channels. The graph legend in the top left of FIG. 9B details the cross-coupling behavior shown by each line in graph **920**. In the graph legend, first radiating element **912a** corresponds to “1”, second

radiating element **912b** corresponds to “2”, third radiating element **912c** corresponds to “3”, and fourth radiating element **912d** corresponds to “4.” For example, the last entry in the graph legend marked “db(S(3,1))_11” refers to the cross-coupling behavior between the third radiating element **912c** and the first radiating element **912a** and is shown by line **924** in graph **920**. It is appreciated that some of the lines shown in the legend may overlap.

(48) A frequency band **922** of approximately 5 GHz band is highlighted between approximately 76 GHz and approximately 81 GHz. As shown in graph **920**, the coupling between the parallel RF radiating antenna elements (i.e., between radiating antenna elements **912a** and **912c** and between radiating antenna elements **912b** and **912d**) is higher compared with the coupling between the orthogonal radiating antenna elements. For example, the cross-coupling shown by line **924** for the internal RF channels associated with the first and third radiating antenna elements **912a** and **912c**, respectively, between 76 GHz and 81 GHz ranges from about -30 dB up to about -25 dB. The cross-coupling between the other internal RF channels associated with the radiating antenna elements **912a-912d** is around -33 dB down to about -47.5 dB between 76 GHz and 81 GHz.

(49) In FIG. **10A**, a layout diagram **1010** is similar to the one shown in layout diagram **910** of FIG. **9A** with the addition of an EBG structure with a plurality of EBG elements **1014** (one labelled in the interest of clarity). As shown, the EBG structure with the plurality of EBG elements **1014** appears inverted (with the supporting column on top of the disc) since the view of the LiP is from the top side view, i.e., the side of the RF IC chip package.

(50) As shown, in some embodiments, the EBG structure with the plurality of EBG elements **1014** does not need to be printed across the entire PCB surface to achieve the advantages described herein (i.e., reducing the cross-coupling between RF channels). For example, as shown in layout diagram **1010**, the EBG structure with the plurality of EBG elements **1014** is arranged to border the internal RF channels associated with the first radiating antenna element **1012a** and the second radiating antenna element **1012b**. In this manner, at least some of the plurality of EBG elements are located between parallel internal RF channels, e.g., the internal RF channels corresponding to first and third radiating antenna elements **1012a** and **1012c**, respectively. By adding the EBG structure with the plurality of EBG elements **1014**, the coupling between the internal RF channels associated with each of the radiating antenna elements **1012a-1012d** is greatly reduced. This reduction in cross-coupling is shown in graph **1020** of FIG. **10B**. The graph legend in the bottom left of FIG. **10B** details the cross-coupling behavior shown by each line in graph **1020**. In the graph legend, first radiating element **1012a** corresponds to “1”, second radiating element **1012b** corresponds to “2”, third radiating element **1012c** corresponds to “3”, and fourth radiating element **1012d** corresponds to “4.” For example, the second entry marked “db(S(3,1))” shows the cross-coupling between the third radiating element **1012c** and the first radiating element **1012a**. As shown, the cross-coupling between first and third radiating antenna elements **1012a** and **1012c** between 76 GHz and 81 GHz drops down to below -75 dB at 79 GHz (compared to about -28 dB at a similar frequency in FIG. **10**) and is shown by line **1022**. The cross-coupling corresponding to the other internal RF channels associated with each of the radiating antenna elements **1012a-1012d** is reduced as well. Accordingly, by implementing an EBG structure as shown and described herein, the cross-coupling between the internal RF channels of the LiP device is reduced, thereby improving the transmission and/or reception signal qualities of the LiP device.

(51) FIG. **11** illustrates a cross-sectional view of a LiP device **1100** in accordance with some embodiments. In some embodiments, the LiP device **1100** corresponds substantially to the LiP device **100** illustrated in FIG. **1** with a difference that the antenna structure in LiP device **1100** is a PCB fanout **1140** as opposed to an antenna array structure **140** as illustrated in FIG. **1**. The PCB fanout **1140** includes vias **1144** in the PCB layer aligned with the internal RF channels **186**, **188**. Thus, the PCB fanout **1140** functions as an RF waveguide to propagate RF signals **126a**, **128a** in the internal RF channels **186**, **188** to and from the RF launchers **126**, **128**, respectively.

(52) FIG. **12** illustrates a cross-sectional view of a LiP device **1200** in accordance with some

embodiments. In some embodiments, the LiP device **1200** corresponds substantially to the LiP device **1100** illustrated in FIG. **11** with a difference that a second EBG structure **1208** is printed on the PCB fanout **1140** side. In some embodiments, the second ENG structure **1208** printed on the PCB fanout **1140** includes a plurality of second EBG elements similar to those illustrated and described in FIG. **6A**, i.e., second RBG structure **1208** includes a plurality of second EBG elements similar to EBG elements **640** shown in FIG. **6A**. By including a second EBG structure **1208** on the PCB fanout side, the band gap of the associated EBG configuration can be more focused on a particular frequency.

(53) In some embodiments, since the radiating antenna structure **1140** in the LiP devices **1100** and **1200** shown in FIGS. **11** and **12** are PCB-based, they may be manufactured using similar techniques used to manufacture PCB **102**. In some embodiments, this reduces the costs of production.

(54) Note that not all of the activities or elements described above in the general description are required, that a portion of a specific activity or device may not be required, and that one or more further activities may be performed, or elements included, in addition to those described. Still further, the order in which activities are listed are not necessarily the order in which they are performed. Also, the concepts have been described with reference to specific embodiments. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the present disclosure as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present disclosure.

(55) Benefits, other advantages, and solutions to problems have been described above with regard to specific embodiments. However, the benefits, advantages, solutions to problems, and any feature(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential feature of any or all the claims. Moreover, the particular embodiments disclosed above are illustrative only, as the disclosed subject matter may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. No limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular embodiments disclosed above may be altered or modified and all such variations are considered within the scope of the disclosed subject matter. Accordingly, the protection sought herein is as set forth in the claims below.

Claims

1. A printed circuit board (PCB) for a launcher in package (LiP) device, the PCB comprising: a first surface that faces an integrated circuit (IC) chip package of the LiP; a second surface that faces an antenna structure of the LiP; and an electromagnetic bandgap (EBG) structure on the second surface, the EBG structure comprising a plurality of EBG elements, wherein: the IC chip package includes a radio frequency (RF) IC chip package with a plurality of RF launchers; the antenna structure includes a plurality of antenna elements, each antenna element of the plurality of antenna elements corresponding to a respective RF launcher of the plurality of RF launchers; the PCB comprises a plurality of vias, each via of the plurality of vias coupling a respective antenna element of the antenna structure with a corresponding RF launcher of the plurality of RF launchers, wherein the plurality of EBG elements borders at least one side of one via of the plurality of vias.
2. The PCB of claim 1, wherein each EBG element of the plurality of EBG elements borders one or more sides of at least one via of the one or more vias.
3. The PCB of claim 2, wherein at least a portion of one of the plurality of EBG elements is located between two vias of the plurality of vias, each of the two vias corresponding to one of two antenna

elements in the antenna structure.

4. The PCB of claim 3, wherein the two antenna elements are parallel to one another.

5. The PCB of claim 1, wherein the plurality of EBG elements protrude from the second surface toward the antenna structure.

6. The PCB of claim 1, wherein the plurality of EBG elements form a periodic pattern with a regular spacing between neighboring EBG elements of the plurality of EBG elements.

7. The PCB of claim 1, wherein each of the plurality of EBG elements comprises a supporting column and a top portion on top of the supporting column.

8. The PCB of claim 7, wherein a height of the supporting column is between 150 micrometers to 350 micrometers.

9. The PCB of claim 7, wherein the top portion is a disc is centered on top of the supporting column and a diameter of the disc is greater than a diameter of the supporting column.

10. The PCB of claim 9, wherein the diameter of the supporting column is between 150 to 500 micrometers and the diameter of the disc is between 500 to 900 micrometers.

11. The PCB of claim 9, wherein the diameter of the disc is at least 1.25 times greater than the diameter of the supporting column.

12. The PCB of claim 9, wherein a thickness of the disc is between 15 to 75 micrometers.

13. A printed circuit board (PCB) for a launcher in package (LiP) device, the PCB comprising: a first surface that faces an integrated circuit (IC) chip package of the LiP, the IC chip package including a radio frequency (RF) IC package with a plurality of RF launchers; a second surface that faces an antenna structure of the LiP, the antenna structure includes a plurality of antenna elements, each antenna element of the plurality of antenna elements corresponding to a respective RF launcher of the plurality of RF launchers; a plurality of vias extending between the first surface and the second surface, each via coupling one of the plurality of RF launchers to a respective one of the plurality of antenna elements; an electromagnetic bandgap (EBG) structure on the second surface, the EBG structure comprising a plurality of EBG elements, each of the plurality of EBG elements borders at least one side of one via of the plurality of vias.

14. The PCB of claim 13, wherein at least a portion of one of the plurality of EBG elements is located between two vias of the plurality of vias, each of the two vias corresponding to one of two antenna elements in the antenna structure.

15. The PCB of claim 13, wherein the plurality of EBG elements protrude from the second surface toward the antenna structure.

16. The PCB of claim 13, wherein the plurality of EBG elements form a periodic pattern with a regular spacing between neighboring EBG elements of the plurality of EBG elements.

17. The PCB of claim 13, wherein each of the plurality of EBG elements comprises: a supporting column; and a top portion on the supporting column, the top portion includes a disc centered on the supporting column; and wherein a diameter of the disc is greater than a diameter of the supporting column.

18. The PCB of claim 17, wherein a height of the supporting column is between 150 micrometers to 350 micrometers.

19. The PCB of claim 17, wherein the diameter of the disc is at least 1.25 times greater than the diameter of the supporting column.

20. The PCB of claim 17, wherein a thickness of the disc is between 15 to 75 micrometers.
